$$(x_1\ x_2\ x_3)\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} a_{11}x_1 + a_{21}x_2 + a_{31}x_3 \\ a_{12}x_1 + a_{22}x_2 + a_{32}x_3 \\ a_{13}x_1 + a_{23}x_2 + a_{33}x_3 \end{bmatrix}$$

$$x_1 + a_{21}x_2 + a_{31}x_3)x_1 + (a_{12}x_1 + a_{22}x_2 + a_{32}x_3)x_2 + (a_{13}x_1 + a_{23}x_2 + a_{33}x_3)x_3$$

$$x_1^2 + a_{21}x_1x_2 + a_{31}x_1x_3 + a_{12}x_1x_2 + a_{22}x_2^2 + a_{32}x_2x_3 + a_{13}x_1x_3 + a_{23}x_2x_3 + a_{33}x_3^2$$

$$a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 + (a_{21} + a_{12})x_1x_2 + (a_{31} + a_{13})x_1x_3 + (a_{32} + a_{23})x_2x_3 \Rightarrow \text{Quadratic form}$$

$a_{11} \Rightarrow$ Coefficient of x_1^2

$a_{22} \Rightarrow$ Coefficient of x_2^2

$a_{33} \Rightarrow$ Coefficient of x_3^2

$a_{21} + a_{12} \Rightarrow a_{12} = a_{21} = \frac{1}{2}$ coefficient of x_1x_2

$a_{31} + a_{13} \Rightarrow a_{13} = a_{31} = \frac{1}{2}$ coefficient of x_1x_3

$a_{32} + a_{23} \Rightarrow a_{32} = a_{23} = \frac{1}{2}$ coefficient of x_2x_3

Three Variables Example

$x_1 \quad x_2 \quad x_3$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & & \\ & & \end{bmatrix}$$

coef of x_1^2	0.5 coef of x_1x_3	0.5 coef of x_1x_2
0.5 coef of x_1x_2	coef of x_2^2	0.5 coef of x_2x_3
0.5 coef of x_1x_3	0.5 coef of x_2x_3	coef of x_3^2